# Forest Engineering

## Science

**May, 2026**

Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

# Forest Engineering (A09991)

| | | | |
|---|---|---|---|
| **Short Title:** | Forest Engineering | | |
| **Faculty** | Science and Computing | | |
| **Department** | Science | | |
| **Cluster** | Forestry | | |
| **Author** | TKENT | | |
| **Credits** | 5 | **Level:** | Intermediate |

## Description of Module / Aims

In this module students will learn about technical solutions to the most important forest operations, i.e. road construction, drainage, ground preparation and planting. During the module students will be required to assess the environmental consequences and health and safety requirements of different technical solutions. Students will also learn about the increasingly important use of wood as a renewable fuel, and the machinery and methods of converting forest residues and small trees to wood particles.

## Programmes

| | | |
|---|---|---|
| FORS-0016 | BSc in Forestry (WD_SFORS_D) | 3 / 6 / M |

## Indicative Content

- Land drainage: types of soil water; principles of drainage; methods of drainage and design of drains; subsoiling, ripping, mole drains, gravel moles, and deep ploughing; environmental concerns in relation to forest drainage
- Forest establishment and maintenance: mechanical site clearing and ground preparation; mechanical planting and vegetation control; pruning and non-commercial thinning
- Forest roads: road and site classification; forest road design; standards for forest roads; road construction; road degradation and maintenance; environmental concerns in relation to forest roads
- Road haulage of forest products: truck configuration; permissible loads; route planning optimisation and agreed routes
- Comminution of wood: characteristics of forest assortments available for comminution; wood chipping, shredding and grinding equipment; wood particle products quality parameters
- Wood energy: characteristics of wood as a fuel; wood burning boiler technologies; quantification and conversion of woodfuel

## Learning Outcomes

*On successful completion of this module, a student will be able to:*

1. Design the drainage requirements, the appropriate method of drainage, and drainage maintenance requirements for a range of land types.
2. Analyse the processes used in forest establishment and maintenance and recommend the appropriate machines and methods for a range of forest sites.
3. Establish the main processes involved in forest road planning, design, construction and maintenance and assess road suitability for different types of timber trucks.
4. Investigate the use of forest resources for wood energy and the machinery and methods used in wood fuel production.
5. Plan forest operations in a manner that complies with best health and safety practice and environmental guidelines.
6. Produce a report and map containing relevant information on site conditions, environmentally sensitive areas and social issues in relation to road building and design a forest road including construction cost.
7. Prepare a forest land development plan, including drainage design, ground preparation and establishment methods and costs.
8. Analyse a wood fuel supply chain and the woodfuel requirements and energy costs of a wood-fired boiler.

## Learning and Teaching Methods

- Lectures
- Fieldwork

## Assessment Methods

| | Weighing | Outcomes Assessed |
|---|---|---|
| Final Written Examination | 50% | 1, 2, 3, 4, 5 |
| Continuous Assessment | 50% | |
| *Project* | 15% | 6 |
| *Project* | 15% | 7 |
| *Project* | 20% | 8 |

## Assessment Criteria

**<40%:** Very limited knowledge and understanding of the subject matter. Lacking in technical ability/skill and understanding to carry out assessments. Failure to meet the objectives of assessments. Unable to carry out or submit independent study and research. Unable to present written work with attention to correct formatting, grammar and spelling.

**40%-49%:** Demonstrate a limited knowledge of the subject matter. Show some technical ability/skill in carrying out and reporting on assessments. Attempt made to meet objectives and carry out independent study and research. Final result lacks balance and arguments undeveloped. Can present written work correctly formatted with minor grammar and spelling errors only.

**50%-59%:** Demonstrate satisfactory general knowledge of the main issues within the subject. Logically and competently carry out practical/professional tasks. Satisfactory handling and awareness in carrying out independent study and research. Reporting and projects adequately structured and referenced.

**60%-69%:** Demonstrate sound knowledge of the subject matter. Show ability to analyse and logically argue in an effective and mature style. Demonstrate initiative and the ability to criticise methods used and appreciation of experimental design when carrying out practical/ professional tasks. Projects and reporting coherent and soundly structured illustrating wide and in depth reading on subject coupled with the proper use of references.

**70%-100%:** Demonstrate authoritative handling of complex material and a highly developed and extensive understanding of the subject. Provide well focused analysis and convincing argument on the subject. Demonstrate imaginative approach, excellent technical ability, awareness, and aptitude in carrying out and reporting on practical/professional tasks. Show initiative, individual thought, and enthusiasm in self study and independent learning.

**Note:**

- In addition to the normal regulations, a minimum of 35% must be achieved in both the final exam and continuous assessment components of the module. In cases where continuous assessment has a practical component attendance of at least 75% in the practicals is mandatory.

## Learning Modes

| Learning Mode | F/T Hours | P/T Hours |
|---|---|---|
| Lecture | 36 | |
| Field Trips | 24 | |
| Independent Learning | 75 | |

## Essential Material


- Anon., 1. _Forestry and Water Quality Guidelines_. Ireland: Forest Service, Department of Agriculture, Food and the Marine, 2000.
- Anon., 2. _Forestry Schemes Manual_. Ireland: Forest Service, Department of Agriculture, Food and the Marine, 2011.
- Collins, K., G. Gallagher, J. Gardiner, E. Hendrick and D. McAree. _Code of Best Forest Practice - Ireland_. Ireland: Forest Service, Department of Agriculture, Food and the Marine, 2000.
- Forest Industry Transport Group. _Managing Timber Transport Good Practice Guide_ by Lyons, J., F. Russell, M. Joyce and G. Devlin.. Dublin, Ireland. 2014.
- Forest Service, Department of Agriculture, Food and the Marine. _Forest Roads Scheme 2014-2020_ Ireland. 2015.
- Irish Forest Industry Chain and Forest Industry Transport Group. _Road Haulage of Round Timber Code of Practice_ by Joyce, M. and E. Daly.. Dublin, Ireland. 2012.
- Kent, T., P. Kofman and E. Coates. _Harvesting wood for energy - Cost-effective woodfuel supp__ly chains in Irish forestry_. Dublin, Ireland: COFORD, 2011.
- Mulqueen, J., M. Rodgers, E. Hendrick, M. Keane and R. McCarthy. _Forest Drainage Engineering - A Design Manual_. Ireland: COFORD, 1999.
- Ryan, T., H. Phillips, J. Ramsey and J. Dempsey. _Forest R__oad Manual: Guidelines for the design, construction and management of forest roads_. Dublin, Ireland: COFORD, 2004.


## Requested Resources

- Lecture Room: Loose Seated